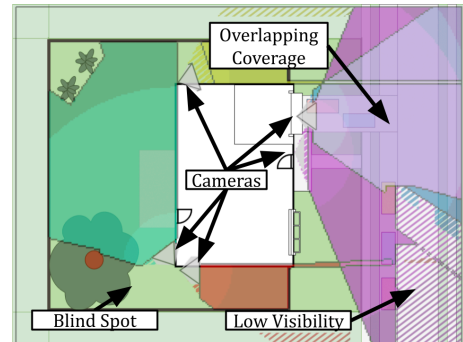


Modeling and Analysis of Security Camera Coverage

Visualization Engineer | Oct 2024 – Apr 2025

Overview:

Developed a reusable custom tool in Blender's Geometry Nodes to efficiently identify optimal security camera placements in outdoor spaces, overcoming limitations of outdated and costly existing software. The tool features real-time camera placement with adjustable simulation resolution, configurable fields of view, multi-camera support, and accurate obstruction modeling using 3D meshes. Future enhancements include streamlining setup, synchronizing Blender's camera system for real-time view simulation, improving node organization, and developing user documentation



Coverage of demonstration property using a common 5-camera setup: four turret cameras and one doorbell camera.

Tools & Skills:

Blender (Geometry Nodes), 3D Modeling and Visualization, Vector Math and Geometric Computation, Line-of-Sight and Occlusion Simulation, Real-Time Performance Optimization

Objective:

Develop a reusable custom tool in Blender's Geometry Nodes for quickly identifying optimal security camera placement in outdoor spaces, addressing limitations in existing software that was outdated, restrictive, or cost-prohibitive.

Design Highlights:

- Real-time camera placement with adjustable simulation resolution settings, enabling fast iteration at low resolution and detailed analysis at high resolution when needed.
- Fully configurable camera field of view, supporting compatibility with a wide range of real-world camera models.
- Multi-camera environment support, allowing simultaneous analysis of complex coverage scenarios.
- Easily add obstructions using any 3D mesh or object, enabling accurate simulation of visual occlusion by structures, foliage, or terrain.

Lessons Learned:

Real-time modification at full resolution proved challenging on consumer-grade hardware, highlighting the necessity of a low-resolution mode to enable rapid adjustments without computational delays. Incorporating small visualization tools significantly improved the intuitive understanding of vector mathematics, making complex operations more accessible. Additionally, using a general-purpose 3D program like Blender enabled straightforward modeling of buildings and obstructions to accurately simulate line-of-sight blockages, demonstrating the flexibility and effectiveness of versatile tools for spatial planning.

Next Steps:

Streamline the project setup process to improve usability and reduce configuration time. Synchronize Blender's camera system with simulation camera objects to enable real-time view simulation. Reorganize and relabel node groups to enhance clarity, maintainability, and ease of modification. Develop documentation to support onboarding and assist users unfamiliar with the tool or its implementation in Blender.